

Practice Quiz: Communication

Part A: Multiple Choice

1. Generalized synchrony in Active Inference refers to: A) Two agents performing the same action at the same time B) A continuous mapping between the state spaces of coupled dynamical systems C) The exact duplication of one agent's generative model in another D) Temporal synchronization of neural oscillations
2. Theory of Mind (ToM) is modeled in Active Inference as: A) A separate cognitive module dedicated to social reasoning B) Hierarchical inference — the agent's model includes a model of the other agent's model C) Mirror neuron activation during social interaction D) An innate, unlearned capacity present from birth
3. According to the Active Inference framework, language primarily functions as: A) A set of labels arbitrarily mapped to objects in the world B) Cues that activate specific patterns in the listener's generative model C) A purely social convention with no cognitive function D) A barrier to direct perception of reality
4. Miscommunication arises in the Active Inference framework when: A) The channel between agents is physically disrupted B) The sender's and receiver's generative models diverge in how they interpret signals C) One agent has a larger vocabulary than the other D) The communication occurs in writing rather than speech
5. Husserl's concept of *Einfühlung* (empathy) relates to Active Inference's view that: A) Understanding another person requires simulating their emotional states B) Understanding requires constructing a model of the other's perspective, not just observing behavior C) Empathy is impossible because Markov Blankets prevent access to internal states D) Emotions are irrelevant to communication
6. Cultural practices (rituals, institutions) function within Active Inference as: A) Obstacles to individual learning B) Long-duration mechanisms for maintaining shared generative models across generations C) Purely symbolic activities with no functional role in inference D) Evidence that Active Inference applies only to individuals, not groups

7. Deception in the Active Inference framework requires: A) The ability to produce sounds
B) Modeling the target's generative model to predict how a false signal will be interpreted
C) Complete knowledge of the target's internal states
D) The deceiver to believe the false information themselves

Part B: Short Answer

1. Explain the Wittgensteinian insight that "the limits of my language are the limits of my world" in Active Inference terms. How does language shape the generative model?
2. Two musicians improvising together gradually develop an intuitive sense of each other's playing. Describe this process as coupled Active Inference: what synchronizes, and how?
3. Construct an argument for or against the claim: "All communication is inherently incomplete because Markov Blankets prevent direct access to another agent's internal states."